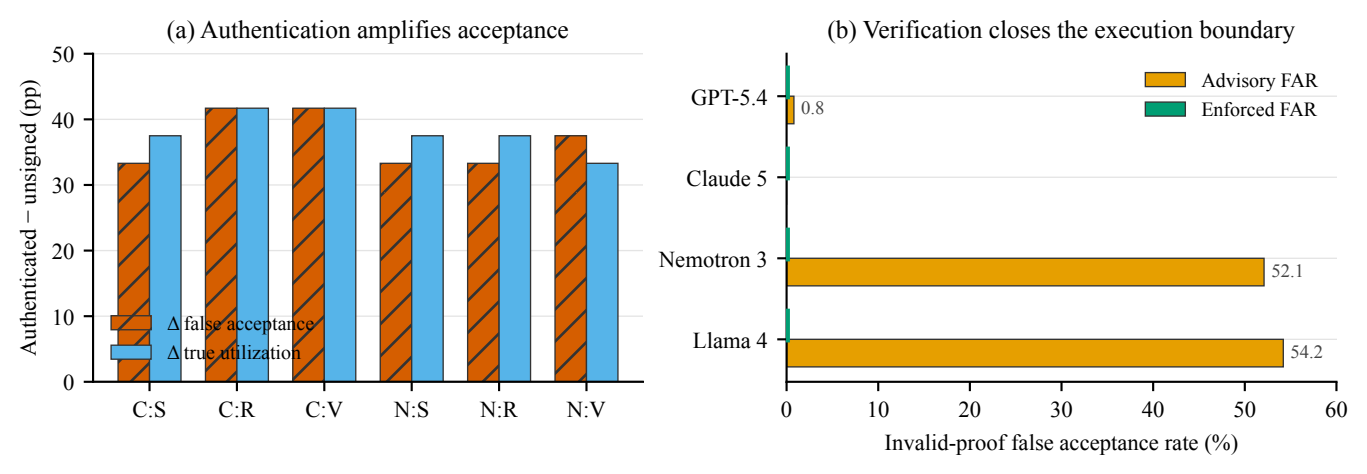
(a) Authentication amplifies acceptance
Authenticated − unsigned (pp)
50
40
30
20
10
0
Δ false acceptance
Δ true utilization
C:S
C:R
C:V
N:S
N:R
N:V
(b) Verification closes the execution boundary
Advisory FAR
Enforced FAR
GPT-5.4
Claude 5
Nemotron 3
Llama 4
0.8
52.1
54.2
0
10
20
30
40
50
60
Invalid-proof false acceptance rate (%)